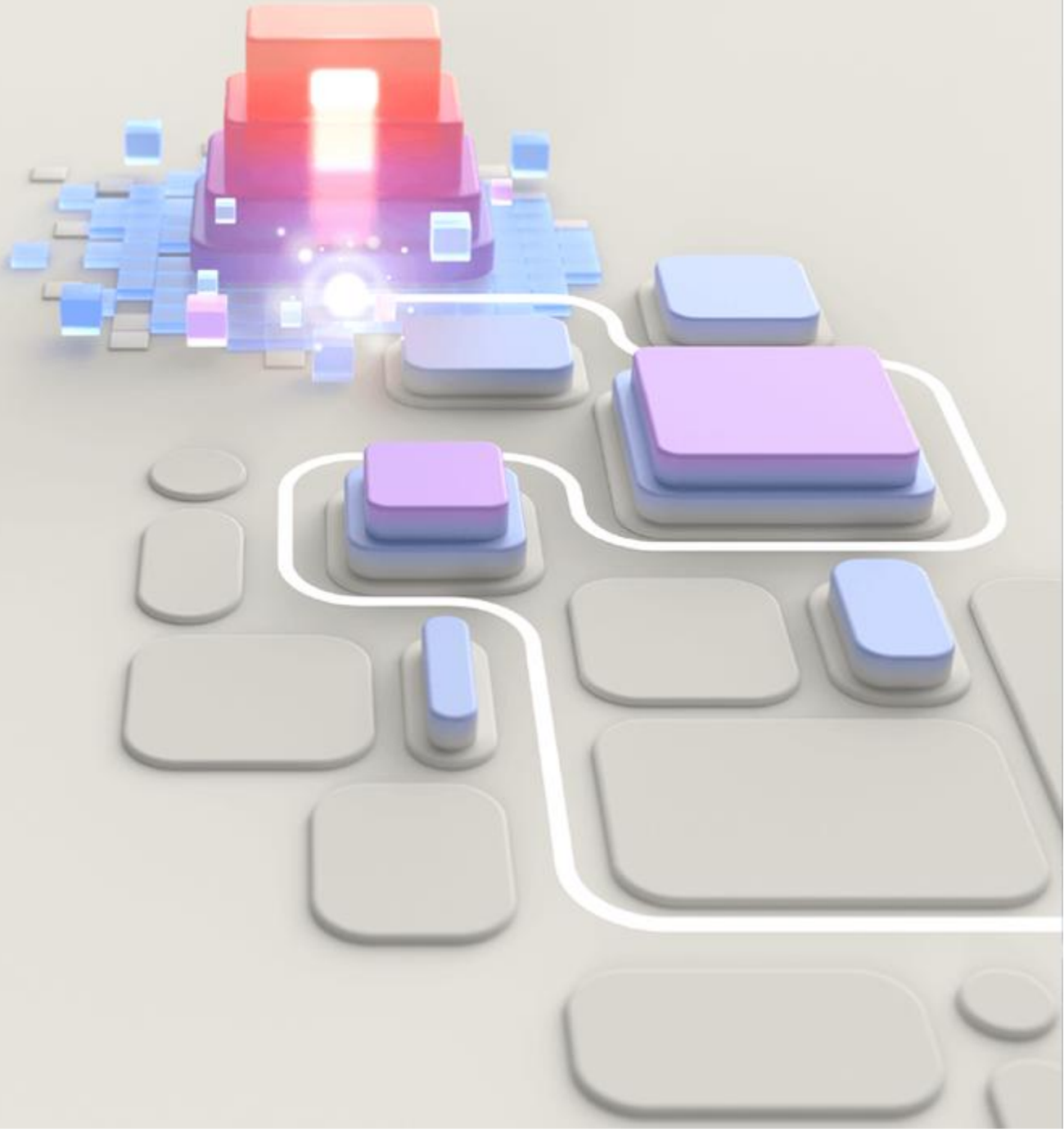
Microsoft

AZ-104

Administer Azure Resources



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AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

Learning Objectives - Azure Resources

- Configure Azure Resources with Tools
  - Microsoft Copilot in Azure
  - Manage services with the Azure portal
  - Introduction to PowerShell
  - Introduction to Bash
- Deploy Azure resources with templates
- Lab 03b - Manage Azure resources by Using ARM Templates

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Configure Azure Resources with the Azure Portal, PowerShell, and the CLI

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Learning Objective – Configure Azure Resources with tools

- Microsoft Copilot for Azure
- Other Administration tools
  - Azure Portal
  - Azure Cloud Shell
  - Azure PowerShell and CLI
- Demonstration – Azure Tools
- Learning Recap

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Review Azure Resource Terminology

A **resource** is simply a single service instance in Azure

A **resource group** is a logical grouping of resources

An **Azure Resource Manager template** is a JSON file that allows you to declaratively describe a set of resources

A **declarative syntax** is what a template uses to state what you intend to create

A **resource provider** is service that supplies the resources you can deploy and manage through Resource Manager

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Review Resource Manager Benefits

Verb Noun PowerShell Module AZ

Get-AzVM

Präfix

-Debug

Tool

az vm list

Postman

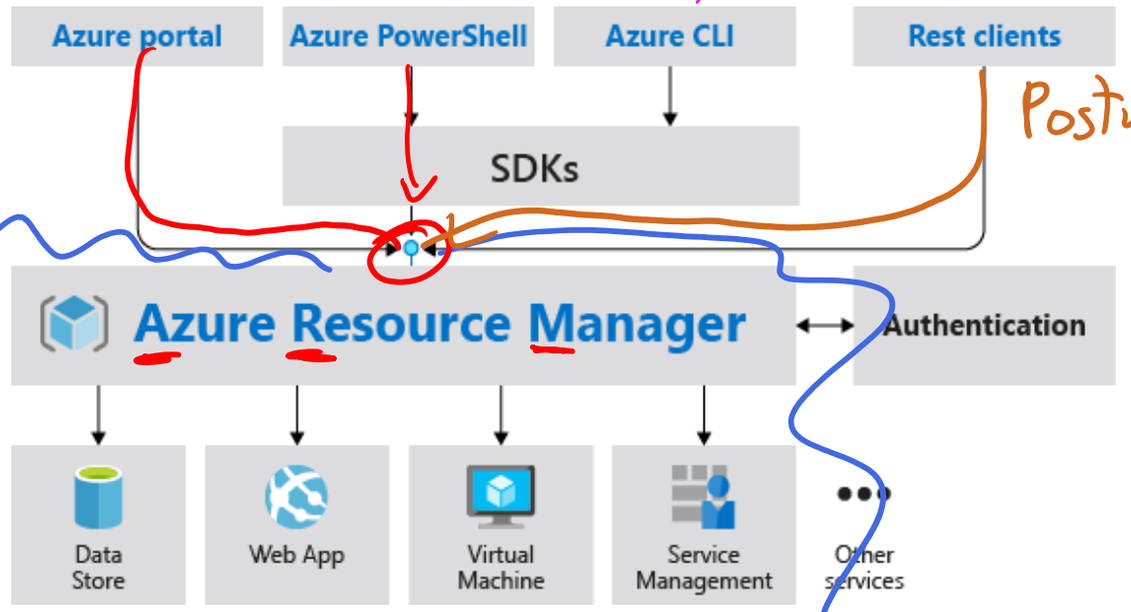
Provides a consistent management layer

Enables you to work with the resources in your solution as a group

Deploy, update, or delete in a single, coordinated operation

Provides security, auditing, and tagging features

Choose the tools and APIs that work best for you



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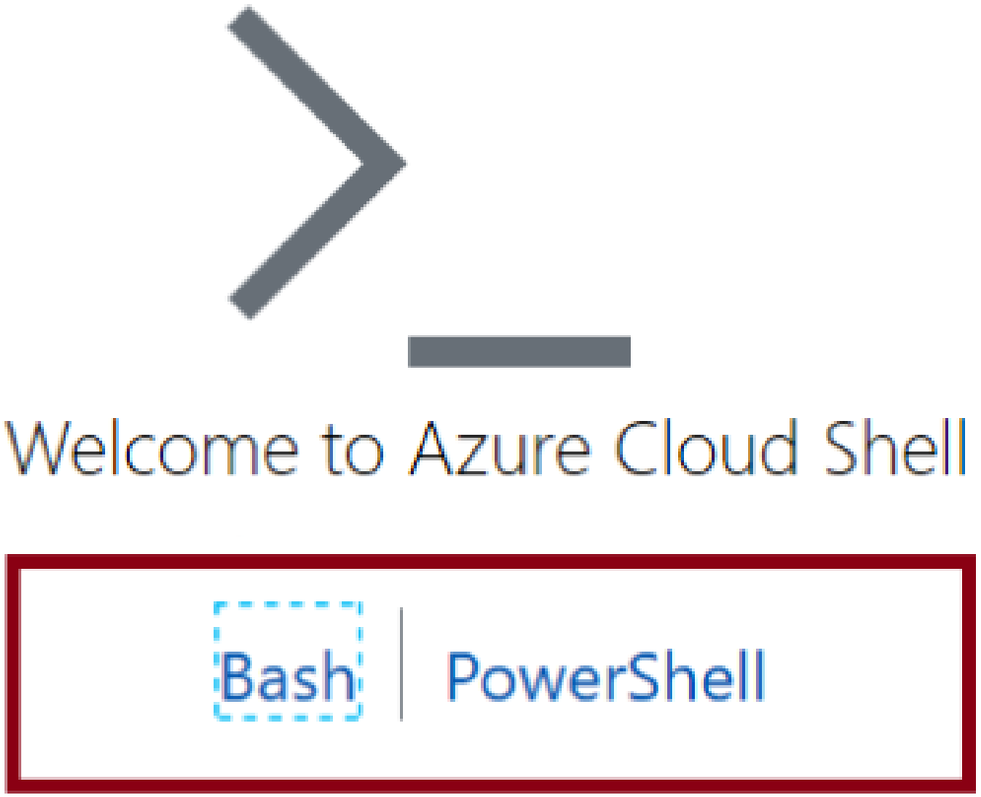
Use Azure Cloud Shell

No Storage (ephemeral)

Storage ($)

- Interactive, browser-accessible shell
- Offers either Bash or PowerShell
- Is temporary and provided on a per-session, per-user basis
- Requires a resource group, storage account, and Azure File share
- Authenticates automatically
- Integrated graphical text editor Code (Monaco)
- Is assigned one machine per user account
- Times out after 20 minutes



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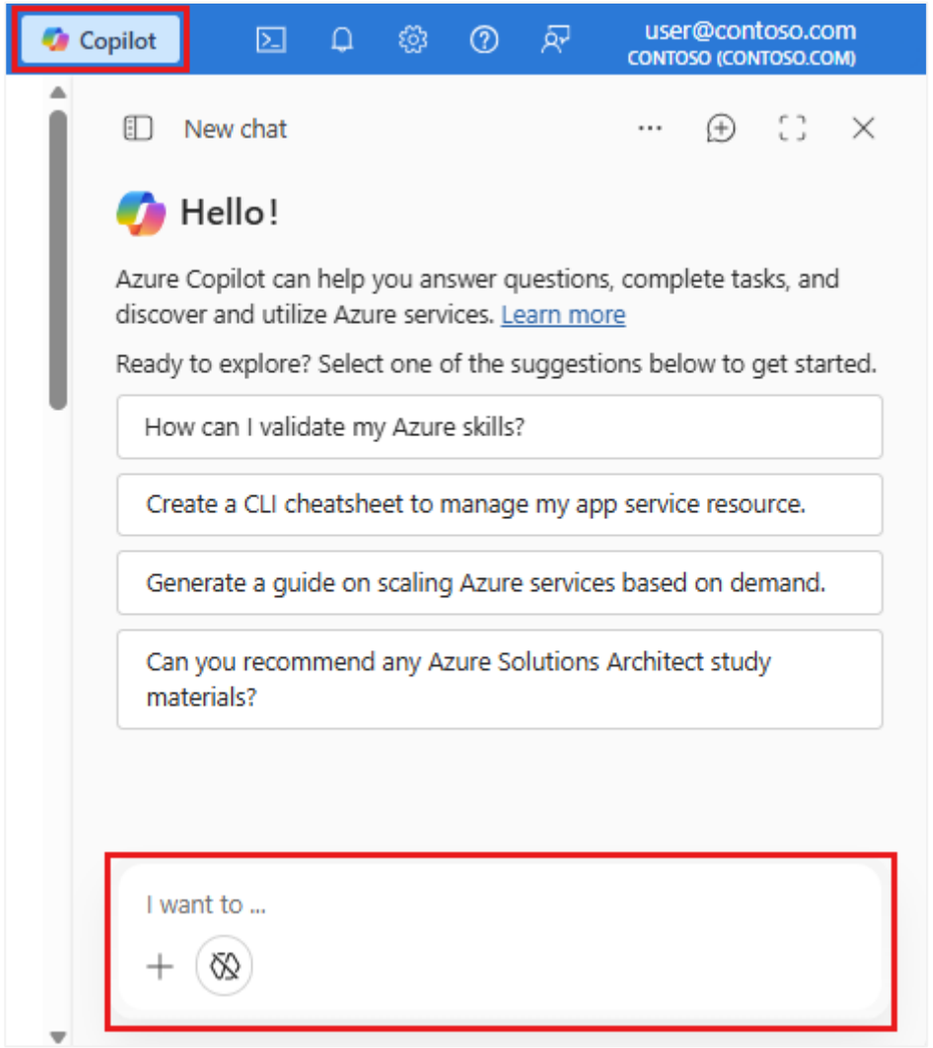
Microsoft Copilot for Azure

Azure Resource Graph

- Access in the Azure portal, through the Azure mobile app, or through AI Shell
- Helps you design, operate, optimize, and troubleshoot your Azure apps and infrastructure
- Performs many basic tasks
  - Deploy and manage virtual machines
  - Design, troubleshoot, and secure networks
  - Generate PowerShell and Azure CLI scripts
  - Analyze, estimate, and optimize costs
  - Discover and deploy workload templates



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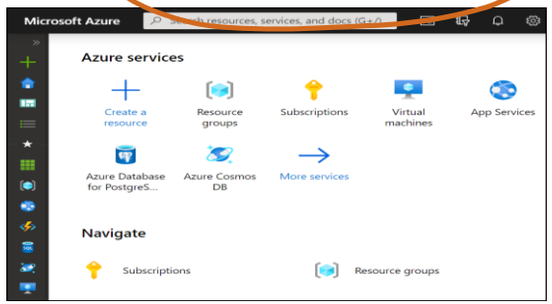
Other Administrator Tools

Powershell

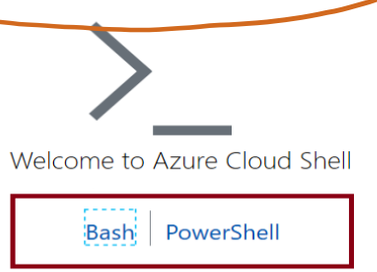
Bash + Powersh

Azure Portal



- View and manage resources
- Visual interface
- Unified hub – training and documentation
- Personalize your experience
- Mobile app
- Access the Cloud Shell
- One-off creation scenarios

Azure Cloud Shell



- Interactive and browser-accessible with file storage
- Offers Bash or PowerShell
- Authenticates automatically
- Provided on a per-session and per-user basis
- Temporary - times out after 20 minutes

Azure PowerShell and CLI

```
az vm restart -g
MyResourceGroup -n MyVm
```

- Command line programs
- Interactive and scripting modes
- Cross-platform
- Good for repeatable deployments
- Familiar coding experience

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Learning Recap – Azure Resources with Tools

Check your knowledge questions and additional study

Discussion Questions

Name at least three tools you can use to manage Azure resources. In what situations would you use each tool?

Reference modules

- Describe features and tools for managing and deploying Azure resources
- Introduction to Azure Cloud Shell
- Introduction to Microsoft Copilot in Azure

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Terraform HCL → ARM API

Bicep → ARM json → ARM API

Deploy Azure resources with templates

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Learning Objective – Deploy Azure resources using templates

- Explore the JSON Template Schema
- Explore the JSON Template Parameters
- Consider Azure Bicep Files
- Demonstration – QuickStart Templates
- Learning Recap

Exam objectives

Automate deployment of resources by using ARM templates or Bicep files

- Interpret an ARM template or a Bicep file
- Modify an existing ARM template
- Modify an existing Bicep file
- Deploy resources by using an ARM template or a Bicep file
- Export a deployment as an ARM template or convert an ARM template to a Bicep file

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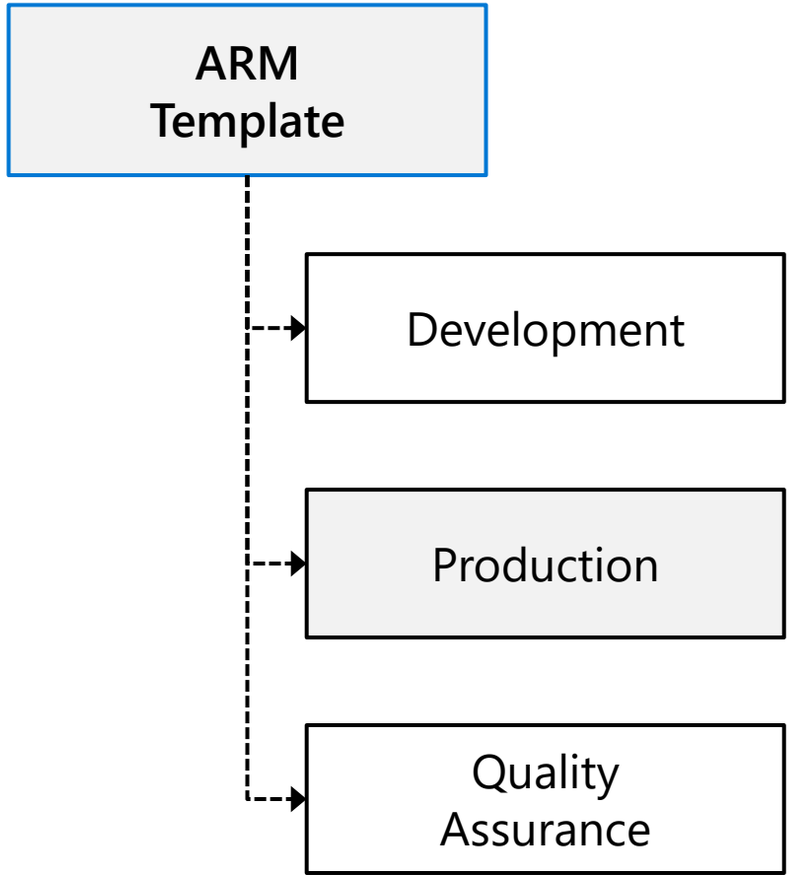
Review ARM Template Advantages

- Improves consistency and promotes reuse
- Reduce manual, error prone, and repetitive tasks
- Express complex deployments
- Express requirements through code
- Provides validation tasks
- Modular and can be linked
- Simplifies orchestration



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Explore the JSON Template Schema

- Defines all the Resource manager resources in a deployment
- Written in JSON
- A collection of key-value pairs
- Each key is a string
- Each value can be a string, number, Boolean expression, list of values, object

```
{
    "$schema":
      "http://schema.management.
      azure.com/schemas/2019-04-
      01/deploymentTemplate.json#",
    "contentVersion": "",
    "parameters": {},
    "variables": {},
    "functions": [],
    "resources": [],
    "outputs": {}
}
```

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Explore the JSON Template Parameters

- Specifies which values are configurable when the template runs
- This example has two parameters: one for a VM's username (adminUsername), and one for its password (adminPassword)

```
"parameters": {
  "adminUsername": {
    "type": "string",
    "metadata": {
      "description": "Username for the VM."
    }
  },
  "adminPassword": {
    "type": "securestring",
    "metadata": {
      "description": "Password for the VM."
    }
  }
}
```

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Consider Azure Bicep Files

Portal

PS New-AzResourceGroupDeployment

CLI az deployment group

- Simpler syntax for writing templates
- Smaller module files you can reference from a main template
- Automatically detect dependencies between your resources
- Visual Studio Code extension with validation and IntelliSense

Bicep file

```
resource storageAccount
'Microsoft.Storage/storageAccounts@
2021-01-01' = {
  name: storageAccountName
  location: location
  tags: {
    displayName: storageAccountName
  }
  kind: 'StorageV2'
  sku: {
    name: 'Standard_LRS'
  } }
```

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Lab - Manage Azure resources by Using ARM Templates

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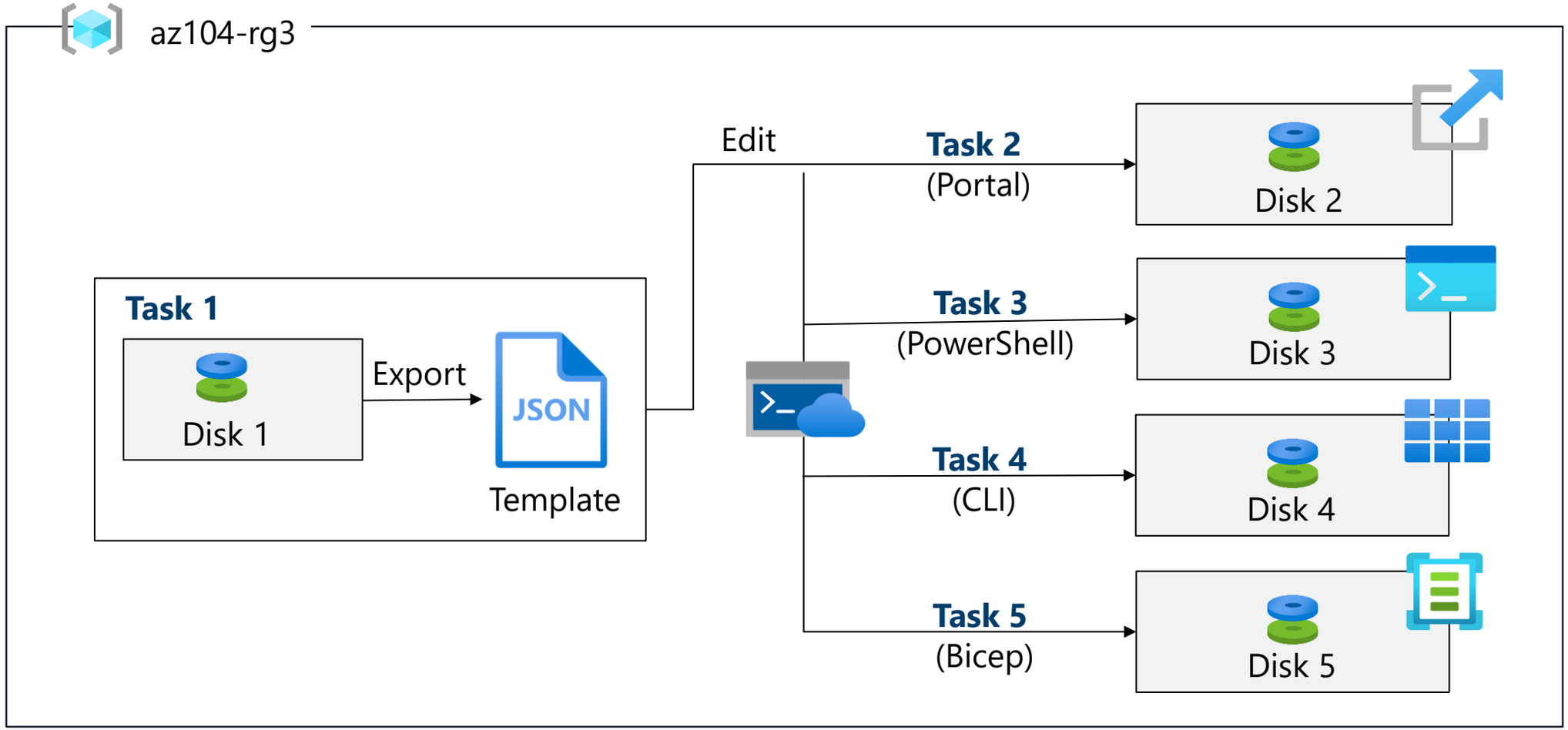
Lab 03 – Architecture diagram



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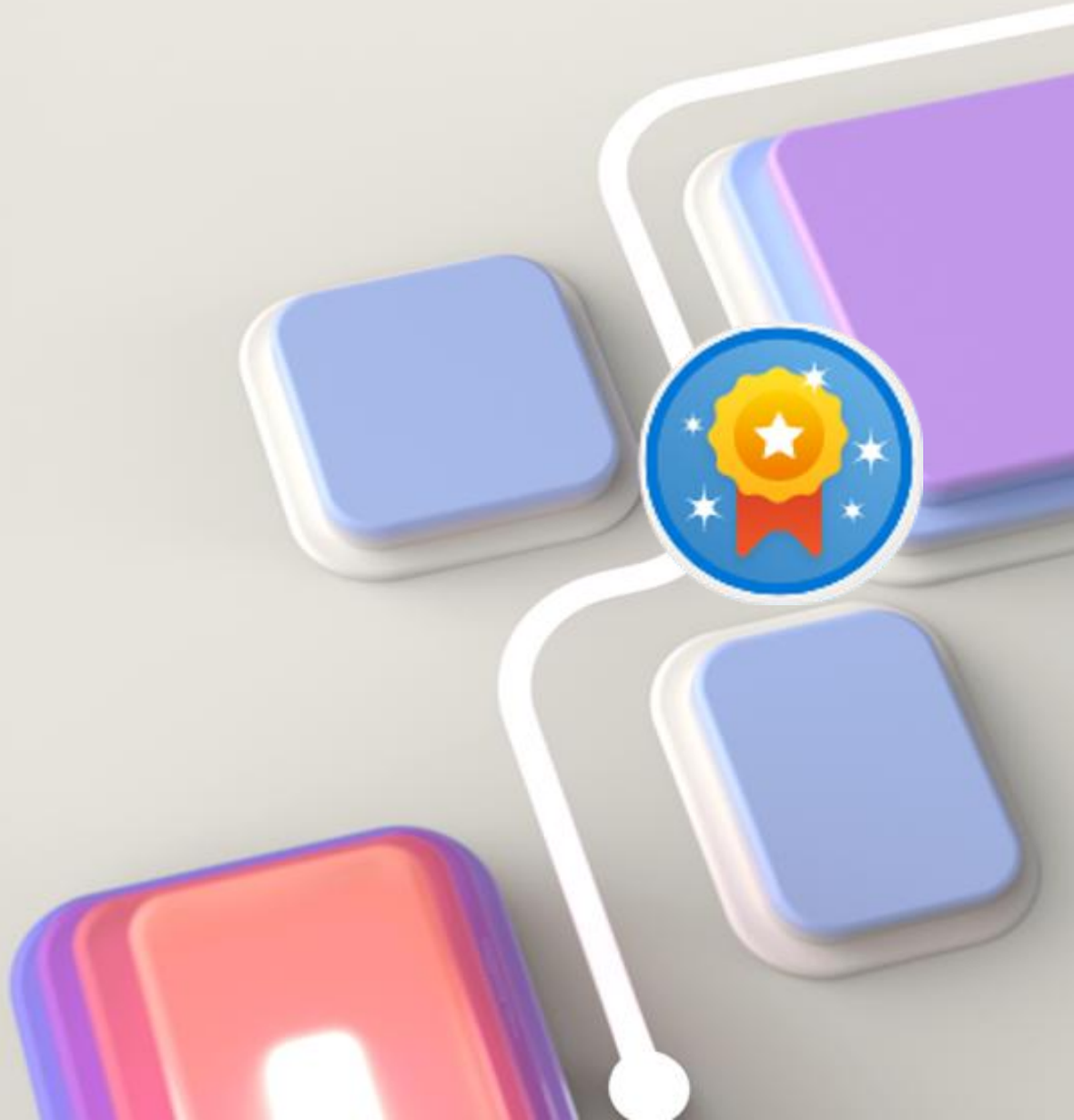
End of presentation



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